

Answer Part 1: Git Use with Engineering Workflow

Review draft — Issue #54. Current local branch: feature/7-lab4-release-evidence. This branch is based on staging commit d5c81a5. It has not been pushed and its PR has not been opened.

Six reviewed feature/documentation PRs have merged into lab4-staging. Each used the required issue branch and requested peer review.

PR	Scope	Review state
#55	Engineering contract	Approved and merged
#56	Actions Taken foundation	Approved and merged
#57	Actions Taken UI	Review change fixed, approved and merged
#58	Ticket workflow	Approved and merged
#59	Role dashboards	Global attention ordering review change fixed, approved and merged
#60	Regression and hardening	Approved and merged

Peer reviewer: Kittithat Disthanakornkun ([@JeffMerry](#)).

PR	Review comment and response
#55, #56, #58	Approved with no blocking corrections.
#57	Requested separate Action list and assignee-lookup failures; implemented independent retry/loading plus regression tests, then approved.
#59	Requested one globally ordered attention top-five query; implemented a combined predicate and ordering regression, then approved.
#60	Confirmed accessibility, responsive/safe-failure coverage and truthful pending database rerun; no blocking change, approved.

Issue and branch list: [#48](#) through [#54](#). Repository structure, setup, and test guidance are in the [README](#). Detailed review links, comments, responses, approvals, and merge outcomes are in the [review record](#); the private Git-flow note is intentionally excluded.

Pending: verify the actual Project #1 Kanban statuses (the current token lacks read:project), complete Issue #54 review, then prepare the separate staging-to-main release PR and record its merge evidence.

Answer Part 2: Spec DD

The Lab 4 contract was reviewed before implementation and included in [PR #55](#). It defines Actions Taken, Ticket transitions and resolution gate, role dashboards and database calculations, migration/seed decisions, API/UI behavior, authorization, concurrency, acceptance criteria, and Product Definition of Done.

- [Engineering specification](#)
- [REST API contract](#)
- [UI specification](#)

Key rules include authenticated actor attribution, Requester ownership, versioned stale-write rejection, append-only Action Taken events, an advisory Requester resolution indication, and a server-enforced resolution gate.

Excluded scope preserved: SLA/escalation, external notifications, inventory/cost/billing/payroll, multi-level approvals, advanced BI/export, multi-tenancy/cloud operations, and unapproved features.

Contract decisions are traceable to the [Lab 4 handout](#); contract layout follows the Lab 3 engineering-document structure.

Answer Part 3: Test DD and Traceability

Test plan and results maps AC-01–AC-21 to API, UI, migration, regression, responsive, accessibility, performance-smoke, and real E2E coverage.

Check	Recorded result
Server suite	19 files / 56 tests passed on Issue #53; reviewer approved PR #60.
Client suite on release branch	15 files / 58 tests passed (2026-10-06).
Client production build on release branch	Passed (2026-10-06).
Client lint on release branch	Passed with one existing react(only-export-components) warning in AuthContext.tsx.
Server TypeScript on release branch	npx.cmd tsc --noEmit passed (2026-10-06).
Real browser E2E	7/7 passed on Issue #53 before the final focus assertion. Current rerun is blocked because PostgreSQL returns ECONNREFUSED and Docker Engine is unavailable.
Mocked browser suite	9/9 displayed as passed, but runner hung after the last test and was interrupted; not counted as a clean pass.

Not yet complete: clean/copy migration recovery verification and the final full test run on the exact main commit.

Answer Part 4: AI Use with Reflection

LLM used: OpenAI Codex. The complete selected prompt record and a clearly labelled reflection draft are in [ai-use.md](#).

Selected prompts used during the work

1. “อ่านรายละเอียดเอกสาร Lab 4 ทั้งหมด ... เขียน specification.md และไฟล์อื่น ๆ ... อ้างอิง handout เป็นหลัก และเน้นข้อห้าม”
2. “สร้างไฟล์ .md ที่ไว้เป็น git flow ... ทำงานเริ่มจาก doc ... ทุก branch merge เข้า lab4-staging ... สร้าง PR และ request @JeffMerry”
3. “แก้ไข issue ... Issue 1: Lab 4 — ... เชื่อม issue กับ kanban board ... และเชื่อม issue กับ branch”
4. “recheck plan ทั้งหมด ... และไฟล์ใน docs\lab-04 ... ตาม SE+Lab+4.md”
5. “ลงรายละเอียดการ prompt ... ไวใส่ที่เดียวตอนท้าย”
6. “ทำงานใน branch ถัดไป ... ยึด docs\lab-04 ... ทำไปทีละ step และ commit”
7. “ดูที่ jeffMerry review มาแล้วแก้ไขด้วย พร้อมตอบกลับ”
8. “ทำงานต่อใน branch นี้ ... อย่าเพิ่ง push ... อ้างอิง format จาก docs\lab-03 ... และข้อกำหนดใน specification.md กับ SE+Lab+4.md”

The quoted lines are concise excerpts; the linked record keeps the selected prompts as used in the conversation and maps each to outcomes. Reflection text is a draft for the student to verify and personalize before submission.

Reflection draft: Using an approved contract and issue-sized branches made changes easier to review. Peer feedback exposed requirements that needed stronger tests, such as globally ordered attention results. I checked outputs and kept mocked browser checks separate from real server/database E2E. Some release evidence still needs a working database, clean/copy migration verification, and confirmation of Project status; I have not represented those as complete.

Answer Part 5: Working IT Staff Dashboard UI

The staff/admin dashboard presents backend-calculated status and priority counts, unassigned and personally owned Tickets, current-user assigned Actions Taken, recent completed work, and recent/high-priority Tickets with Queue drill-downs.

Evidence: [reviewed dashboard PR #59](#); [database-backed API tests](#) compare returned metrics to Prisma count predicates; responsive screenshots are in [staff-dashboard evidence](#).

These tests verify database-aligned values and role restrictions. A separate captured SQL transcript is not included in this draft; the API test's database queries are the recorded oracle.

Answer Part 6: Working Actions Taken UI

Actions Taken are distinct from Public Comments and Internal Notes. Staff can create, assign, edit, start, complete, and cancel actions; completion records result and actor. Follow-up and attachment notes are retained. Requesters see read-only actions only on owned Tickets.

The [UI PR #57](#) and [hardening PR #60](#) record review. API/component tests cover validation, inactive assignees, idempotent retries, stale conflicts, role limits, audit, and retained history. Screenshots: [Actions Taken desktop/tablet/mobile](#).

The real browser lifecycle passed in the earlier 7/7 run. The newly added focus-restoration assertion passed its client component test; the full real browser suite still needs rerunning when PostgreSQL is available.

Answer Part 7: Working Ticket Workflow

The backend enforces the Ticket transition matrix, optimistic versions, and the resolution prerequisites: completed work with a result, no planned/in-progress actions, and no active required follow-up. Requester “appears resolved” remains advisory. Action and Ticket histories preserve audit order.

Evidence: [Ticket workflow PR #58](#), [workflow API tests](#), and [Action Taken API/audit tests](#). Administrators retain approved Lab 4 operational access and User Management; Requester ownership and private-note rules remain enforced.

Answer Part 8: Working Requester Dashboard and Final Regression

The Requester dashboard scopes counts and records to the authenticated owner and links to My Tickets or owned Ticket detail. API tests compare calculations with database predicates and cover UTC boundaries, globally ordered attention, empty results, and legacy unresolved timestamps.

Labs 1–3 API, component, and real-browser coverage is included in the full server/client suites and 7/7 real E2E result recorded in [tests.md](#). Lab 2 archived requester-selector browser tests are excluded because authentication replaced that selector; this exclusion and the non-clean mocked runner result are disclosed there.

Final regression on main remains pending. This report does not claim the main branch contains the Lab 4 release.

Answer Part 9: Zen Green UI, Responsive Design, Accessibility, and Final Polish

UI conventions and keyboard/responsive requirements are in the [UI specification](#). Twelve screenshots cover Staff Dashboard, Requester Dashboard, and Actions Taken at desktop, tablet, and mobile. Real responsive browser tests check page horizontal overflow.

Checklist	Evidence/status
Zen Green colors, labelled states, visible focus	Client style tests pass.
Semantic labels and validation associations	Component/style tests and PR #60 review.
Actions dialog keyboard trap and live feedback	Earlier real E2E and UI tests.
Focus restoration after the source button is removed	Component test passes; new full real E2E assertion awaits database-backed rerun.
Desktop/tablet/mobile screenshots and no page overflow	12 screenshots and responsive E2E pass.
Kanban Issues #48–#54 all Done; final main tests and PDF links verified	Pending; Project API scope and release integration are not yet available/complete.

Scope check: no SLA, external notification, inventory/cost/billing/payroll, multi-level approval, advanced BI/export, multi-tenant/cloud, or other unapproved feature was added.

Draft status: This is a review copy on the local release-evidence branch, not the final submission copy. Replace pending items only after verifying them against the exact final main commit and Project board.